

Question #1

$$T(n) = T(n/2) + 1$$

$$T(1) = 1$$

$$a = 1$$

$$b = 2$$

$$f(n) = 1 = n^0$$

$$\log_b(a) = \log(1)/\log(2) = 0$$

$$f(n) = O(n^0)$$

$$n^{\log_b(a)} = n^0$$

$$T(n) = O(n^0 * \log n)$$

$f(n)$ is better than $\log n$

$$T(n) = O(\log n)$$

Question #2

$$T(n) = T(n-1) + n^2$$

$$T(1) = 1$$

$$T(n) = T(n-1) + n^2$$

$$T(n) = T(n-2) + (n-1)^2 + n^2$$

$$T(n) = T(n-3) + (n-2)^2 + (n-1)^2 + n^2$$

Sum of Square

$$T(n) = n^3$$